

HSE Intelligence

KPI Formula Verification Report

Organisation:	Alex Carry / WindTech (org_id=4)
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Database:	hse_db (MySQL, localhost:3306)
Data Range:	Jan 2024 - Dec 2025
Total Incidents:	53
Total Near Misses:	120
Total Safety Walks:	27 (in 90-day window)
Total Man Hours:	622,370
Report Date:	17-07-2026

1. Predictive Injury Risk Score

VERIFIED | Dashboard Value: 58.33% (down 5.95%)

Formula

Score = (weight_sum / (count x 3)) x 100

Severity Weight Mapping

Severity	Weight
Critical / Significant	3
High / Major	2
Medium / Moderate	1
All others (Minor, Serious, Low)	0.5

Date Window

Latest incident date: 2025-12-19
Current period: 2025-09-20 to 2025-12-19 (last 90 days, end excluded)
Previous period: 2025-06-22 to 2025-09-20

Current Period Incidents (8 records)

ID	Date	Type	Severity	Weight
187	2025-09-25	Damage	Significant	3
201	2025-10-09	Injury	Serious	0.5
182	2025-10-13	Environmental	Significant	3
171	2025-10-22	Environmental	Minor	0.5
207	2025-10-30	Near-miss	Minor	0.5
184	2025-10-31	Environmental	Significant	3
181	2025-11-08	Damage	Significant	3
218	2025-11-14	Near-miss	Serious	0.5

Calculation

weight_sum = 3 + 0.5 + 3 + 0.5 + 0.5 + 3 + 3 + 0.5 = 14
count = 8
Score = (14 / (8 x 3)) x 100 = (14/24) x 100 = 58.33%

Trend Calculation

Previous period (7 incidents): weight_sum=13.5, count=7
Previous Score = (13.5 / (7x3)) x 100 = (13.5/21) x 100 = 64.29%
Trend = 58.33 - 64.29 = -5.95% (improvement)

2. TRIR (Total Recordable Injury Rate)

VERIFIED | Dashboard Value: 3.53

Formula

TRIR = (Recordable Injuries x 200,000) / Man Hours

Data from DB

Recordable Injuries = COUNT(*) FROM incidents
WHERE incident_type = 'Injury' AND date <= 2025-12-19
Result: 11

Man Hours = SUM(actual_hours_worked) FROM shift_schedule
Result: 622,370

All Recordable Injuries (11)

ID	Date	Severity
174	2024-01-20	Lost Time
206	2024-05-20	Serious
183	2024-07-03	Significant
194	2024-07-12	Significant
193	2024-09-02	Minor
175	2025-01-15	Minor
215	2025-05-08	Serious
205	2025-06-07	Significant
180	2025-06-22	Significant
208	2025-08-23	Significant
201	2025-10-09	Serious

Calculation

TRIR = (11 x 200,000) / 622,370

= 2,200,000 / 622,370

= 3.53

Why 200,000?

OSHA standard: 100 workers x 40 hrs/week x 50 weeks = 200,000
Normalizes rate to 'per 100 full-time workers per year'

3. LTIFR (Lost Time Injury Frequency Rate)

VERIFIED | Dashboard Value: 1.61

Formula

$$LTIFR = (Lost\ Time\ Injuries \times 1,000,000) / Man\ Hours$$

Data from DB

Lost Time Injuries = COUNT(*) FROM incidents
WHERE incident_type='Injury' AND severity='Lost Time'
Result: 1 (ID=174, date 2024-01-20)

Man Hours = 622,370

Calculation

$$\begin{aligned} LTIFR &= (1 \times 1,000,000) / 622,370 \\ &= 1,000,000 / 622,370 \\ &= 1.61 \end{aligned}$$

Why 1,000,000?

ILO international standard.
Lost Time Injuries are rare, so a larger multiplier gives meaningful numbers.

4. Near Miss Ratio

VERIFIED | Dashboard Value: 10.9 : 1

Formula

$$Ratio = Near\ Miss\ Count / Recordable\ Injuries : 1$$

Data from DB

Near Miss Count = COUNT(*) FROM near_misses WHERE organisation_id=4
Result: 120

Recordable Injuries = 11 (same as TRIR)

Calculation

$$\begin{aligned} Ratio &= 120 / 11 = 10.909... \\ \text{Rounded to 1 decimal} &= 10.9 \\ \text{Display: } &10.9 : 1 \end{aligned}$$

Interpretation

For every 1 actual injury, 10.9 near misses were reported.
Based on Heinrich's Safety Triangle concept.
Higher ratio = better (proactive reporting culture)
Industry benchmark: 10:1 to 30:1 is considered good.

5. Audit Readiness Score

VERIFIED | Dashboard Value: 75.6% / Needs Attention

Formula

Score = (AVG(compliance_rating) / 5) x 100

Labels

Score Range	Label
>= 80%	Ready
>= 60% and < 80%	Needs Attention
< 60%	Not Ready

Date Window

Latest safety walk date: 2025-12-30
Window: last 90 days = 2025-10-01 to 2025-12-30
Source: safety_walks.compliance_rating (scale 1-5)

Safety Walk Compliance Ratings (27 records)

ID	Date	Rating
893	2025-10-04	5
794	2025-10-07	3
883	2025-10-08	5
756	2025-10-08	5
931	2025-10-14	4
846	2025-10-14	3
726	2025-10-15	4
797	2025-10-16	3
740	2025-10-18	3
857	2025-10-23	5
876	2025-11-04	2
853	2025-11-18	2
749	2025-11-22	5
801	2025-11-23	5
741	2025-11-24	3
821	2025-12-01	3
882	2025-12-03	4
924	2025-12-03	3
729	2025-12-04	3
778	2025-12-09	3
725	2025-12-13	5
890	2025-12-16	5

792	2025-12-24	4
733	2025-12-26	4
713	2025-12-28	4
731	2025-12-29	5
739	2025-12-30	2

Calculation

Sum of all ratings = 5+3+5+5+4+3+4+3+3+5+2+2+5+5+3+3+4+3+3+3+5+5+4+4+4+5+2 = 102

Count = 27

AVG = 102/27 = 3.7778 -> rounded = 3.78

Score = (3.78 / 5) x 100 = 75.6%

75.6 >= 60 -> Label: 'Needs Attention'

6. DART Rate (Days Away, Restricted, Transferred)

VERIFIED | Dashboard Value: 0.32

Formula

$$DART = (Lost\ Time\ Injuries \times 200,000) / Man\ Hours$$

Calculation

Lost Time Injuries = 1
Man Hours = 622,370
 $DART = (1 \times 200,000) / 622,370 = 0.32$

7. LTISR (Lost Time Injury Severity Rate)

VERIFIED | Dashboard Value: 1.61

Formula

$$LTISR = (Lost\ Days \times 1,000,000) / Man\ Hours$$

Calculation

Lost Days = SUM(days_away) WHERE incident_type='Injury' AND severity='Lost Time'
Result: 1
Man Hours = 622,370
 $LTISR = (1 \times 1,000,000) / 622,370 = 1.61$

8. FAR (Fatal Accident Rate)

VERIFIED | Dashboard Value: 0

Formula

$$FAR = (Fatalities \times 100,000,000) / Man\ Hours$$

Calculation

Fatalities = COUNT(*) WHERE severity='Fatal' = 0
Man Hours = 622,370
 $FAR = (0 \times 100,000,000) / 622,370 = 0$

9. Exposure Index (Intelligence-Based)

VERIFIED | Dashboard Value: 68%

Formula

Exposure Index = round(AVG(compliance_rating) x 20)

Why x 20?

compliance_rating is on a 1-5 scale.

To convert to percentage: multiply by 20 (since $5 \times 20 = 100\%$).

This is equivalent to $(\text{avg} / 5) \times 100$ but expressed as $\text{avg} \times 20$.

Data from DB

Source: safety_walks.compliance_rating (ALL records, no date filter)

Organisation: org_id = 4

$\text{AVG}(\text{compliance_rating}) = 3.4$

Calculation

Exposure Index = $\text{round}(3.4 \times 20)$
= round(68)
= 68%

Note

Despite the name 'Exposure Index', on the Dashboard this gauge measures safety walk compliance (higher = better compliance).

The People page has a separate 'Worker Exposure Index' with a different formula based on recent incidents + near misses.

10. Competency Coverage (Intelligence-Based)

VERIFIED | Dashboard Value: 74%

Formula

Competency Coverage = round((CAPA_Completed / CAPA_Total) x 100)

Data from DB

Source: capa_actions table (org-wide, no date filter)

Organisation: org_id = 4

CAPA Completed (status = 'Completed') = 31

CAPA Total = 42

Calculation

capa_completion_rate = $\text{round}((31 / 42) \times 100, 1) = 73.8$

Frontend: $\text{Math.round}(73.8) = 74\%$

Interpretation

This metric shows what percentage of Corrective/Preventive Actions (CAPA) have been completed. Higher = better organizational competency to address identified safety issues.

74% means 31 out of 42 total CAPA actions are resolved.

PEOPLE PAGE (/users) KPIs

11. Competency Coverage % (People Page)

VERIFIED | Dashboard Value: 81%

Formula

$$\text{competency_pct} = \text{round}((\text{total_emp} - \text{flagged}) / \text{total_emp} * 100)$$

What is 'flagged'?

An employee is 'flagged' (not competent) if:

- 1. They reported an incident where root_cause or root_cause_category contains 'train' (training-related root cause), OR
- 2. They are assigned a CAPA action where root_cause_addressed contains 'train'

Data from DB

total_employees = 150

Flagged from incidents (reported_by where root_cause LIKE '%train%'):

21 unique employees

IDs: 20,34,35,57,60,61,73,81,82,86,90,91,92,93,98,121,123,131,136,138,143

Flagged from CAPA (responsible_person_id where root_cause_addressed LIKE '%train%'):

9 unique employees

IDs: 20,36,37,52,69,84,117,133,149

Union (set merge): 29 unique (ID 20 is common in both)

Calculation

$$\begin{aligned} \text{competency_pct} &= \text{round}((150 - 29) / 150 * 100) \\ &= \text{round}(121 / 150 * 100) \\ &= \text{round}(80.67) \\ &= 81\% \end{aligned}$$

Labels

Score Range	Tone	Label
>= 80%	Green	Excellent
>= 60% and < 80%	Amber	Good
< 60%	Red	Needs Improvement

12. Worker Exposure Index (People Page)

VERIFIED | Dashboard Value: 18%

Formula

$$\text{exposure} = \text{round}(\text{min}(100, (\text{incidents} + \text{near_misses}) / \text{employees} * 100))$$

Date Window

activity_anchor = MAX(latest_incident, latest_near_miss)
= MAX(2025-12-19 17:24, 2025-12-19 09:00)
= 2025-12-19 17:24:00
cutoff_90 = 2025-12-19 17:24:00 - 90 days = 2025-09-20 17:24:00

Data from DB

recent_incidents (>= cutoff_90) = 9
recent_near_misses (>= cutoff_90) = 18
total_employees = 150

Calculation

exposure_index = round(min(100, (9 + 18) / 150 * 100))
= round(min(100, 27/150 * 100))
= round(min(100, 18.0))
= 18%

Labels

Score Range	Tone	Label
> 30%	Red	High Risk
>= 10% and <= 30%	Amber	Medium Risk
< 10%	Green	Low Risk

18% falls in Amber zone = 'Medium Risk'

13. Supervisor Safety Score (People Page)

VERIFIED | Dashboard Value: 66%

Formula

score = round(AVG(compliance_rating) / 5 * 100) for supervisors

Logic

1. Find all roles where safety_signatory = 'Yes'
Result: Plant Manager, Safety Manager, Operations Manager, Dept Supervisor
2. Find safety walks where inspector's role has safety_signatory = 'Yes'
(JOIN safety_walks -> employees -> roles WHERE safety_signatory='Yes')
3. Average their compliance_rating

Data from DB

AVG(compliance_rating) for supervisor inspectors = 3.32

Total employees = 150

Calculation

supervisor_score = round(3.32 / 5 * 100)
= round(66.4)
= 66%

Label: 'Needs Coaching' (< 70)

Labels

Score Range	Label
>= 90%	Highly Effective
>= 70%	Effective
< 70%	Needs Coaching

14. Behaviour Breakdown (People Page)

Formula

percentage = round(category_count / total * 100)

Data from DB

Safe = safety_walks WHERE issues_found = 0 => 41

At-Risk = safety_walks WHERE issues_found > 0 => 184

Near Miss = near_misses count => 120

Total = 41 + 184 + 120 = 345

Calculation

Category	Count	Formula	Result
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Safe	41	$\text{round}(41/345*100)$	12%
At-Risk	184	$\text{round}(184/345*100)$	53%
Near Miss	120	$\text{round}(120/345*100)$	35%

15. Fatigue Trend (People Page)

Formula

Per week: normal = min(hours, 8), overtime = max(0, hours - 8)

Logic

Window: latest shift date (2025-12-31) back 10 weeks
= 2025-10-23 to 2026-01-01

For each shift record in this window:
- normal hours = min(actual_hours_worked, 8)
- overtime hours = max(0, actual_hours_worked - 8)

Aggregated by week number (0-9), displayed as stacked bar chart showing normal vs overtime hours per week.

16. Training Expiry (People Page)

Formula

For each employee with induction_date:
For each training program with expiry_months > 0:
Calculate next_due = induction_date + N*expiry_months until > today
Classify: Expired / Due <30 Days / Due <90 Days

Data from DB

Training programs for org=4: 0 (none exist)
Therefore: Expired=0, Due <30 Days=0, Due <90 Days=0
expiring_soon_count = 0

Summary - All KPIs Verified

KPI	Formula	DB Result	Dashboard	Status
Injury Risk Score	$(\text{weight}/\text{count} \times 3) \times 100$	58.33%	58.33%	PASS
TRIR	$(\text{injuries} \times 200\text{K}) / \text{hours}$	3.53	3.53	PASS
LTIFR	$(\text{LTI} \times 1\text{M}) / \text{hours}$	1.61	1.61	PASS
Near Miss Ratio	$\text{NM} / \text{injuries}$	10.9:1	10.9:1	PASS
Audit Readiness	$(\text{avg_rating} / 5) \times 100$	75.6%	75.6%	PASS
DART	$(\text{LTI} \times 200\text{K}) / \text{hours}$	0.32	0.32	PASS
LTISR	$(\text{days} \times 1\text{M}) / \text{hours}$	1.61	1.61	PASS
FAR	$(\text{fatal} \times 100\text{M}) / \text{hours}$	0	0	PASS
Exposure Index	$\text{avg_compliance} \times 20$	68%	68%	PASS
Competency Coverage	$(\text{capa_done} / \text{total}) \times 100$	74%	74%	PASS
--- PEOPLE PAGE ---	---	---	---	---
Competency %	$(\text{emp-flagged}) / \text{emp} \times 100$	81%	81%	PASS
Worker Exposure	$(\text{inc} + \text{nm}) / \text{emp} \times 100$	18%	18%	PASS
Supervisor Score	$\text{avg_sup_rating} / 5 \times 100$	66%	66%	PASS
Behaviour-Safe	$\text{safe} / \text{total} \times 100$	12%	12%	PASS
Behaviour-AtRisk	$\text{atRisk} / \text{total} \times 100$	53%	53%	PASS
Behaviour-NearMiss	$\text{nm} / \text{total} \times 100$	35%	35%	PASS

Data Sources

Table	Key Fields	Records Used
incidents	incident_type, severity, incident_date_time, days_away	53 total
shift_schedule	actual_hours_worked	622,370 total hours
near_misses	event_date_time	120 total
safety_walks	compliance_rating, inspection_date_time	All + 27 in window
capa_actions	status, responsible_person_id	42 total (31 done)

Multiplier Standards Reference

Multiplier	Standard Body	Meaning
200,000	OSHA (US)	Rate per 100 full-time workers/year
1,000,000	ILO (International)	Rate per million hours worked
100,000,000	UK/International	Rate per 100 million hours worked